

Breaking Knowledge Boundaries: Cognitive Distillation-enhanced Cross-Behavior Course Recommendation Model

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Abstract

Online Course Recommendation (CR) stands as a promising educational strategy within online education platforms, with the goal of providing personalized learning experiences for learners and enhancing their learning efficiency. Existing CR methods focus on modeling learners' learning needs from their historical course interactions by adopting general recommendation techniques, but fail to consider the shifts in course preferences caused by cognitive states. While Cognitive Diagnosis (CD) techniques are adept at tracking cognitive states' evolution via mining learner-exercise interactions and benefit the CR task, it is non-trivial to integrate CD and CR properly due to several challenges, including accurate diagnosis, divergent task objectives, and inconsistent data magnitude. To address these challenges, we propose a Cognitive Distillation-enhanced Cross-Behavior Course Recommendation model (C^3Rec), which aims to transfer the knowledge of learners' cognitive states to enhance the CR task. Specifically, for accurate diagnosis, we introduce a dual-granularity cognitive diagnosis module to capture learner representations at both coarse and fine granularities, thereby achieving a comprehensive construction of learners' cognitive states. For divergent task objectives, we design a cross-behavior course recommendation module to jointly profile the dynamic course preferences from two temporal interleaved learning behaviors, achieving the seamlessly semantic alignment between these two tasks. For inconsistent data magnitude, we introduce a triple-stage distillation mechanism to exploit cognitive state features as prior knowledge, enhancing the CR task by further profiling learners' course preferences. Experimental comparisons with multiple state-of-the-art methods on two real-world educational datasets demonstrate the effectiveness of our model.

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CCS Concepts

• **Information systems** → **Recommender systems**; **Personalization**; • **Applied computing** → **E-learning**.

Keywords

Course Recommendation, Cognitive Distillation, Cross-Behavior

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1 Introduction

In recent years, the surge in online learning can be attributed to the rapid growth of online education platforms with Massive Open Online Courses (MOOCs). However, the exponential growth of MOOCs in real-world educational scenarios causes the issues of information overload and learning disorientation [13, 25], which makes it challenging for learners to efficiently identify contents that meet their learning needs from vast online educational resources. To tackle these challenges, Course Recommendation (CR) has emerged as a promising educational strategy within online MOOC platforms. By mining the learning needs of learners from their historical course interactions, CR aims to identify high-quality courses that mostly match learners' learning needs, thereby enhancing their learning efficiency and offering personalized learning experiences. In this background, CR has garnered significant research attention.

Existing CR methods can be divided into two categories: Course-grained methods and Concept-grained methods. Course-grained methods, such as FISM [20] and DARL [24], characterize learners' course preferences by aggregating embeddings of historically interacted courses. However, these methods only consider the coarse-grained information contained in the course title texts without taking into account the fine-grained knowledge concept information related to the courses. Given the limitations of course-grained methods, concept-grained methods focus on enhancing CR tasks by integrating fine-grained knowledge concept information to better profile learning needs. For example, Zhang et al. [45] and Sun et

al. [34] emphasize constructing a knowledge concept association network for courses and mapping learners' course-learning behaviors into the course space to enrich learners' representations, thus achieving better recommendation results.

Although concept-grained methods show good results in the CR task, they ignore the impact of learners' cognitive states. In the context of course recommendation, learners' learning needs are influenced by both course preferences and cognitive states [9]. Course preferences govern learners' engagement tendencies, while cognitive states reflect learners' current learning competencies and future learning needs [12]. Therefore, to quantify the cognitive states features, we introduce Cognitive Diagnosis (CD), which can precisely assess learners' mastery of specific knowledge concepts by analyzing historical exercise interactions [36]. Representative works in CD research, such as DINA [7] and IRT [30], have demonstrated effectiveness in assessing learners' learning capabilities and latent learning needs. Consequently, in this paper, we attempt to develop an effective recommendation model equipped with the CD technique for online CR.

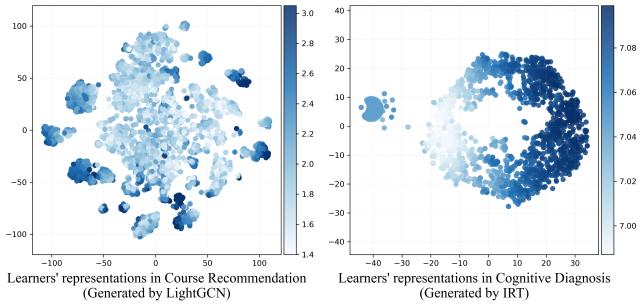


Figure 1: A toy example using the t-SNE method to visualize the gap of learners' representation between cognitive diagnosis and course recommendation respectively. The horizontal and vertical axes in the figure represent non-dimensional and low-dimensional embedding space dimensions, and the relative distances between representations reflect their correlations. Regions in the figure where representations are more concentrated and darker in color indicate that these learners have stronger similarities in their representations.

However, integrating cognitive diagnosis with course recommendation presents three significant challenges. 1) Most of the recent CD methods overlook the importance of fine-grained knowledge concepts information, which leads to the inability to achieve precise modeling. 2) The divergent objectives caused by two different types of interactions result in a gap in learners' representations. As shown in Figure 1, for the same group of learners, the clustered distributions of representations obtained from the two tasks are significantly different, making it difficult to align the two representations when integrating the two tasks. 3) The inconsistent data magnitude between two different types of interactions leads to data imbalance during cross-task fusion, thereby diminishing the performance of course recommendation. To bridge this gap, we innovatively introduce knowledge distillation from transfer learning [37]. This technique can establish a cross-domain alignment bridge by regarding the CD model as a teacher network, where high-order cognitive features can guide the student recommendation model through a distillation mechanism.

In this paper, we propose a **C**ognitive Distillation-enhanced **C**ross-Behavior **C**ourse **Rec**ommendation model (C^3Rec), which aims to enhance online course recommendations by integrating cognitive diagnosis and knowledge distillation. Specifically, for accurate diagnosis, we introduce a dual-granularity cognitive states modeling module to model learners' cognitive states at both exercise and knowledge concept granularities. For divergent task objectives, we build a learner-centric cross-behavior learning graph to integrate learners' two temporal interleaved learning behaviors from two different tasks and implement joint modeling of cognitive states and course preferences via an enhanced cross-graph encoder. For inconsistent data magnitude, we introduce a triple-stage joint learning cognitive distillation mechanism that dynamically merges cognitive state features output by the diagnosis module, thereby exploiting cognitive state features as prior knowledge to enhance the recommendation module. We demonstrated the effectiveness of our method on two real-world MOOC datasets through comparative experiments with state-of-the-art approaches, ablation study, hyperparameter experiment, and case study.

Our key contributions are as follows:

- We design a Cognitive Distillation-enhanced Cross-behavior Course Recommendation model (C^3Rec). We profile learning needs by jointly integrating course preferences and cognitive states, overcoming traditional methods' reliance on single-granularity.
- We propose a dual-granularity cognitive diagnosis module based on coarse-grained exercises and fine-grained knowledge concepts information, thereby achieving accurate diagnosis.
- We propose a cross-behavior course recommendation component to capture the collaborative influence of course and exercise interactions, with a triple-stage joint learning cognitive distillation component to jointly model cross-task cognitive representations.
- Comprehensive experiments on two MOOC datasets with multiple baselines demonstrate C^3Rec's superiority over state-of-the-art methods in Recall and NDCG metrics.

2 Related work

2.1 Course recommendation

The rapid development of online education platforms has driven in-depth research on personalized CR. In recent years, the emergence and evolution of deep learning technologies have accelerated the advancement of course recommendation methods [15]. Deep learning-based approaches autonomously learn high-dimensional feature representations from large-scale educational data, enabling efficient course filtering and recommendation, significantly improving performance [8]. Depending on whether knowledge concept modeling is considered, existing methods can be broadly categorized into **Course-grained methods** and **Concept-grained methods**.

Course-grained methods focus on mining course preferences information from learners' historical course interactions. For instance, Fan et al. [10] propose an interpretable course recommendation method using multi-attention mechanisms to model learning needs from learner-course interactions. Lin et al. [24] and Zhang et al. [46] design hierarchical reinforcement learning framework to adaptively model learner representations, while Wang et al. [38] introduce hypergraph neural networks to enrich learner representations by modeling inter-learner relationships, and Sun et al. [34] construct a course prerequisite graph reflecting learning principles to model learning needs. While these methods effectively leverage rich information from learning behaviors, they ignore the impact of fine-grained knowledge concept information on course preferences.

Concept-grained methods achieve good performance by integrating fine-grained knowledge concepts information. A representative example is KGAN [45], which constructs a heterogeneous graph linking courses and knowledge concepts to iteratively mine latent preferences. CLGADN [26] enhances course recommendations by examining the knowledge granularity associations of courses, and CKFN [27] cascades keyword-level and title-level knowledge graphs to capture fine-grained features in courses. These methods aim to model course preferences from knowledge concept granularity but overlook the critical influence of other learning behaviors, such as exercise-answering. The most relevant work proposed by Ban et al. [2] integrates knowledge tracing with course recommendation through a multi-task learning framework to enhance recommendations with knowledge states, yet falls short in effectively addressing the cross-task representation gap. In contrast, our method innovatively bridges the cross-task representation gap by leveraging knowledge distillation techniques, thereby enabling dynamic collaborative optimization between cognitive states and course preferences.

2.2 Cognitive Diagnosis

Cognitive diagnosis aims to evaluate learners' mastery of knowledge concepts embedded in exercises by analyzing their historical answer performance [35]. Early cognitive diagnosis research stems largely from the realm of educational psychology, with representative works including IRT [30], MIRT [4], and DINA [7]. The IRT model uses logistic functions to simulate the relationship between cognitive proficiency and exercise interactions. MIRT, as an extension of IRT, represents learners more comprehensively across multiple dimensions. DINA employs discrete-continuous product functions to model learners' exercise-answering behaviors. Traditional cognitive diagnosis methods assess learners' cognitive states by modeling their responses to exercises but have simplistic model structures.

With the widespread application of deep learning technologies, neural network-based cognitive diagnosis methods achieve more refined modeling of learners' cognitive states by integrating complex network architectures with traditional models. Wang et al. [36] first reformulate cognitive diagnosis models through neural networks to capture fine-grained learner-exercise interactions. Bi et al. [3] introduced a universal Bayesian meta-learning cognitive diagnosis framework, enhancing the reliability of diagnostic results through Bayesian hierarchical modeling, while Zhao et al. [47] incorporated diffusion models into cognitive diagnosis to eliminate the impact of learners' anomalous behaviors. These methods directly map learners' responses to exercises to their mastery of corresponding knowledge concepts. In contrast, our method utilizes a joint analysis of two distinct hierarchical relationship graphs to collaboratively model learners' cognitive states at the granularity of both exercises and knowledge concepts.

2.3 Knowledge Distillation

Knowledge distillation is an efficient, powerful technique enabling knowledge transfer across different models, data domains, and tasks [37]. It mitigates both the impact of data distribution disparities and task constraints between the source-domain model and target-domain model, transferring knowledge from a larger teacher model to a smaller student model. Based on the type of distilled knowledge, existing methods fall into two categories: **Logit-based methods** and **feature-based methods** [14].

Logit-based methods aim to transfer knowledge by aligning the student network's predictions with the teacher network's logits. Hinton et al. [18] pioneered this idea, using temperature-scaled softmax outputs from the teacher's logits as soft labels and minimizing the Kullback-Leibler divergence between student predictions and soft labels. Hedge et al. [17] enhanced distillation accuracy by designing a sparse/smooth regularizer. Xie et al. [41] utilized both labeled and unlabeled data to prevent overfitting/underfitting during knowledge transfer. Meng et al. [29] proposed a critical learning strategy where the student only adopts teacher predictions when they are correct. Zhou et al. [48] introduced a dual semantic approximation framework to emphasize prior knowledge for improved distillation.

While logit-based methods effectively transfer output-level knowledge, they ignore the rich feature-level information in the teacher network's hidden layers. **Feature-based methods** align the student network's hidden layers with the teacher network, using intermediate features as supervision. Romero et al. [33] first proposed hint-based training, where the teacher's hidden layer outputs guide student learning. Aguilar et al. [1] aligned internal representations across models for abstract knowledge transfer. Yang et al. [43] introduced auxiliary modules to extract and transfer features at each training stage. In this paper, given the divergent output logic between cognitive diagnosis and recommendation tasks, we adopt feature-based distillation to transfer cognitive state features. We align the cognitive state representations from the cognitive diagnosis module with learner representations in the recommendation task, enhancing course recommendations through cognitive-level guidance.

3 Problem Formulation

In this article, we define the learner set as $U = \{u_1, u_2, \ldots, u_{|U|}\}$, the course set as $C = \{c_1, c_2, \ldots, c_{|C|}\}$, the exercise set as $E = \{e_1, e_2, \ldots, e_{|E|}\}$, and the knowledge concept set as $K = \{k_1, k_2, \ldots, k_{|K|}\}$, where$|U|$, $|C|$, $|E|$, and $|K|$ denote the number of learners, courses, exercises, and knowledge concepts, respectively. Each course c_i is associated with at least one knowledge concept $k_i \in K$. Similarly, each exercise e_i is linked to at least one $k_i \in K$. The historical interaction of learner u_i with learning resources (courses or

exercises) is represented as $u_i = \left\{(s_1, r_{i1}), (s_2, r_{i2}), \ldots, \left(s_j, r_{ij}\right)\right\}$, where s_j denotes the interacted resource, either a course c or exercise e, and $r_i j \in 0, \omega, 1$ captures the interaction outcome. For course recommendation, $r_i = 1$ if u_i has interacted with the target course; otherwise, $r_i = 0$. Similarly, for cognitive diagnosis, $r_i = 1$ if u_i has interacted with and correctly answered the target exercise, $r_i = \omega$ if the answer is incorrect, and $r_i = 0$ if no interaction occurs. Here, ω is a hyperparameter where $\omega \in [0, 1]$. Our goal is to infer learners' cognitive states and course preferences from their historical interactions to predict their potential future course engagements.

4 Method

In this section, we will introduce the C^{3}Rec model in detail. As shown in Figure 2, the overall framework of C^{3}Rec is comprised of two modules: a dual-granularity cognitive diagnosis module which collaboratively model learners' cognitive states at the granularity of exercises and knowledge concepts, and a cognitive-distilled cross-behavior course recommendation module which implement joint modeling of cognitive states and course preferences by fusing knowledge distillation.

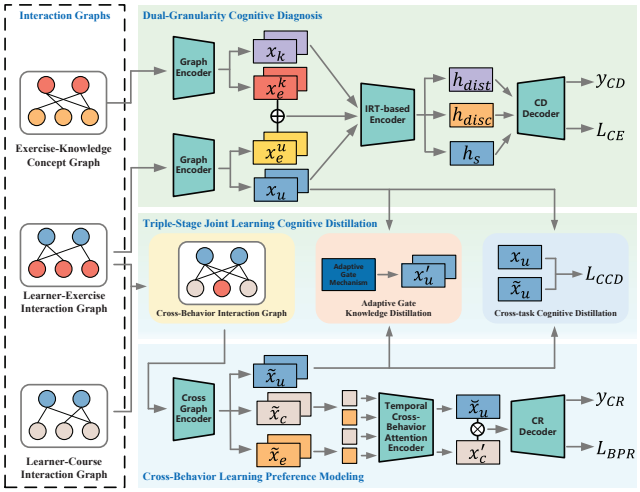


Figure 2: The overall framework of the C^{3}Rec model.

4.1 Dual-Granularity Cognitive Diagnosis

Considering that learners' cognitive levels are often embedded in information at multiple-granularity, we collaboratively model learners' cognitive states from relationship graphs in two different granularities. Specifically, we first construct a learner-exercise interaction graph $G_{ue} = \{v_{ue}, \varepsilon_{ue}\}$, where the node set $v_{ue} = U \cup E$ includes learners U and exercises E, and the edge set ε_{ue} represents learner-exercise interactions. A connected edge exists between learner u_i and exercise e_j in graph G_{ue} if $r_{ij} = 1$ or ω. We define matrix $M^{ue} \in R^{|U|\times|E|}$ to represents the weights of ε_{ue}, with $M^{ue}_{ij} = r_{ij}$. To capture fine-grained knowledge concept information embedded in exercises, we further construct an exercise-concept relation graph $G_{ek} = \{v_{ek}, \varepsilon_{ek}\}$, where the node set $v_{ek} = E \cup K$ includes exercises E and knowledge concepts K, and the edge set ε_{ek} denotes exercise-concept containment relationships. Similarly, we define the edge weight matrix $M^{ek} \in R^{|E|\times|K|}$. If exercise e_i contains knowledge concept k_j, $M^{ek}_{ij} = 1$; otherwise, $M^{ek}_{ij} = 0$. Following the lightweight setting in [16], we employ an extended GCN encoder to propagate multi-hop information on G_{ue} and G_{ek}, respectively, thereby obtaining representations for learners, exercises, and cognitive proficiency. Specifically, for graph G_{ue}, the information propagation process at the l-th layer is defined as:

$$\begin{cases} x_u^{(l)} = \sum\limits_{e\in N_u} \frac{1}{\sqrt{|N_u|}\sqrt{|N_e|}} x_u^{(l-1)}, \\ x_e^{u(l)} = \sum\limits_{u\in N_e} \frac{1}{\sqrt{|N_e|}\sqrt{|N_u|}} x_e^{(l-1)}, \end{cases} \tag{1}$$

where $x_u^{(l)} \in R^d$ and $x_e^{u(l)} \in R^d$ denote the propagated representations of learners and exercises at the l-th layer, with d being the embedding size. $x_u^{(0)}$ and $x_e^{(0)}$ are initialized as learnable parameters at the beginning of training. N_u and N_e represent the neighbor nodes of learners and exercises in graph G_{ue}. Furthermore, we concatenate the representations from each layer at the l-th layer to derive the final representation, enabling awareness of neighbor information across varying hops. The aggregation process is formulated as:

$$x_u = sigmoid\left(\sum_{l=0}^{L} x_u^{(l)}\right),\ x_e^u = sigmoid\left(\sum_{l=0}^{L} x_e^{u(l)}\right), \tag{2}$$

where x_u denotes the final learner representation on graph G_{ue}, x_e^u represents the knowledge concept representations, and $sigmoid(\cdot)$ denotes the activation function. Similarly, for graph G_{ek}. For x_k, we perform the same propagation operation:

$$\begin{cases} x_e^{k(l)} = \sum\limits_{k\in N_e} \frac{1}{\sqrt{|N_e|}\sqrt{|N_k|}} x_e^{k(l-1)}, \\ x_k^{(l)} = \sum\limits_{e\in N_k} \frac{1}{\sqrt{|N_k|}\sqrt{|N_e|}} x_k^{(l-1)}, \end{cases} \tag{3}$$

$$x_e^k = sigmoid\left(\sum_{l=0}^{L} x_e^l\right),\ x_k = sigmoid\left(\sum_{l=0}^{L} x_k^l\right), \tag{4}$$

where x_e^k denotes the knowledge concept-grained exercise representation on graph G_{ek}, and x_k represents the knowledge concept representation. Then, to capture learners' cognitive states through learners' responses to exercises, we adopt the Item Response Theory model (IRT) [30] as the foundation for our cognitive diagnosis module. Specifically, to achieve precise diagnosis, following the original setting in IRT, we introduce a learning factor encoder to encode learners' presentations into three key factors: **student factor** h_s, which represents learner; **knowledge distinction factor** h_{dist}, which indicates the distinction of each knowledge concepts included by one exercise; and **exercise discrimination factor** h_{disc}, which indicates the capability of exercise to differentiate learners with different cognitive states. Here, the learner representation x_u is directly mapped to h_s. For h_{dist}, we generate it by aggregate x_k through average pooling based on graph G_{ek}:

$$h_{dist} = \frac{1}{|N_e|} \sum_{k\in N_e} x_k, \tag{5}$$

where N_e denotes the set of knowledge concepts contained in a single exercise. For h_{disc}, we generate it by concatenating two exercise representations from dual-granularity, followed by nonlinear

fusion via a fully connected layer. The formulation is as follows:

$$h_{disc} = (W_1 \times (x_e^u || x_e^k) + b_1), \tag{6}$$

where W_1 and b_1 are trainable parameters. Subsequently, we pass the processed features through a CD decoder to obtain the final prediction of learners' future performance, which contains an MLP layer and a prediction layer:

$$f_0 = (x_u - h_{dist}) \times h_{disc}, \tag{7}$$

$$f_1 = sigmoid(W_2 \times f_0 + b_2), \tag{8}$$

$$f_2 = sigmoid(W_3 \times f_1 + b_3), \tag{9}$$

$$y_{CD} = sigmoid\left(W_4 \times f_2 + b_4\right), \tag{10}$$

where W_2, W_3, W_4, b_2, b_3, and b_4 are trainable parameters. Finally, we adopt the cross-entropy [28] between the predicted value y_{CD} and the ground-truth label $\hat{y}_{CD}$ as the loss function for the cognitive diagnosis task:

$$L_{CE} = -\sum_i \left(\hat{y}_{CD}\, logy_{CD} + (1 - \hat{y}_{CD})\ logy_{CD}\right). \tag{11}$$

After training, x_u performs as the output of the cognitive diagnosis module, characterizing the learners' evolving cognitive state features and providing cognitive-level enhancement signals for course recommendations.

4.2 Cognitive-Distilled Cross-Behavior Course Recommendation

In real-world learning activities, learners' exercise-answering and course-learning behaviors are deeply intertwined, leading to dynamic coupling effects between answer performance and course preferences. To address this, we construct a cognitive-distilled cross-behavior course recommendation module, comprising two components: A cross-behavior learning preference modeling component that captures learners' course preferences in two temporal interleaved learning behaviors, combined with a triple-stage joint learning cognitive distillation component that transfers cognitive features to optimize course recommendations.

Considering the divergent task objectives between cognitive diagnosis and course recommendation, we attend to modeling course preferences to integrate learner-course and learner-exercise interactions across tasks. Specifically, we first build a learner-course interaction graph $G_{uc} = \{v_{uc}, \varepsilon_{uc}\}$, where the node set $v_{uc} \in U \cup C$ includes learners U and courses C, and the edge set ε_{uc} represents learner-course interactions. A connected edge exists between learner u_i and course c_j in graph G_{uc} if $r_{ij} = 1$. The edge weight matrix $M^{uc} \in R^{|U|\times|C|}$ of edge set ε_{uc} records learner-course interactions, with $M_{ij}^{uc} = r_{ij}$. To model the collaborative impact of two types of interactions to course preferences, we construct a learner-centric cross-behavior learning graph $G_{us} = \{v_{us}, \varepsilon_{us}\}$ by concatenating the two interaction graphs along the learner dimension. Node set $v_{us} \in U \cup C \cup E$ containing learners U, courses C and exercises E, with edge set ε_{uc} covering both learner-course and learner-exercise. The edge weight matrix of G_{us} is formulated as:

$$M^{us} = (M^{uc})^T || (M^{ue})^T, \tag{12}$$

where the edge weight matrix $M^{us} \in R^{|U|\times(|C|+|E|)}$ is constructed by vertically concatenating M^{uc} and M^{ue}, with $M_{ij}^{us} = r_{ij}$. For the cross-behavior graph G_{us}, consistent with the cognitive diagnosis module, we employ the same GCN encoder for information propagation, ultimately obtaining joint representations of learners, courses, and exercises:

$$\begin{cases} \tilde{x}_u^l = \sum\limits_{s\in N_u} \frac{1}{\sqrt{|N_u|}\sqrt{|N_k|}} e_s^{l-1}, \\ \tilde{x}_c^l = \sum\limits_{u\in N_c} \frac{1}{\sqrt{|N_c|}\sqrt{|N_u|}} e_u^{l-1}, \\ \tilde{x}_e^l = \sum\limits_{u\in N_e} \frac{1}{\sqrt{|N_e|}\sqrt{|N_u|}} e_u^{l-1}, \end{cases} \tag{13}$$

$$\widetilde{x}_u = \sum_{l=0}^{L} \tilde{x}_u^l, \widetilde{x}_c = \sum_{l=0}^{L} \tilde{x}_c^l, \widetilde{x}_e = \sum_{l=0}^{L} \tilde{x}_e^l, \tag{14}$$

Where $\widetilde{x}_u$, $\widetilde{x}_c$, and $\widetilde{x}_e$ are representations of learner, course, and exercise in the cross-behavior space. Then, to establish the relationship between courses and exercises, we define a course-exercise relation matrix $M^{ce} \in R^{|C|\times|E|)}$, where $M_{ij}^{ce} = 1$ if course c_i is associated with exercise e_j. Based on M^{ce}, we aggregate exercise representations $\widetilde{x}_e$ into fine-grained exercise-level course representations $\widetilde{x}_c^e$ through feature aggregation:

$$\tilde{x}_c^e = \frac{1}{|N_c^e|} \sum_{e\in N_c^e} \widetilde{x}_e, \tag{15}$$

where N_c^e denotes the set of exercises corresponding to a single course. Finally, we concatenate the course representations $\widetilde{x}_c$ and $\widetilde{x}_c^e$ from two distinct levels to obtain the final course representation:

$$x_c^{'} = W_5 \left(\widetilde{x}_c \middle\| \tilde{x}_c^e\right) + b_5, \tag{16}$$

where W_5 and b_5 are trainable parameters. Moreover, to align cognitive state representations from cognitive diagnosis tasks with learner representations, we designed a temporal cross-attention encoder leveraging the contextual modeling capabilities of self-attention to capture cross-behavior temporal dependencies between course-learning and exercise-answering behaviors. For learner u_i, we first integrate course representations x_c' and exercise representations $\widetilde{x}_e$ to construct the temporal interaction sequence $s_i = [x_1, x_2, \ldots, x_n]$, where x_n denotes the embedding of the n-th interacted courses or exercises. Next, we obtain the positional encoding of s_i through computation:

$$p_t{}^{(i)} = \begin{cases} \sin\left(w_k \times t\right), & i = 2k \\ \cos\left(w_k \times t\right), & i = 2k+1 \end{cases}, \tag{17}$$

We then inject temporal information into s_i via learnable positional encoding vectors:

$$s_i^t = s_i + p_t, \tag{18}$$

where $w_k = \frac{1}{10000^{2k/d}}$, and d represents the embedding size. Finally, for the learners' cross-behavior temporal representation s_i^t, we feed it into a self-attention encoder to obtain learner representations that capture dynamic shifts in learning needs. The Query, Key, and Value of the self-attention encoder are computed as:

$$\begin{cases} Q = W_Q S_i^t \\ K = W_K S_i^t \\ V = W_V S_i^t \end{cases}, \tag{19}$$

where W_Q, W_K and W_V are learnable weight matrices. The scaled dot-product attention operation is then applied:

$$x_u^s = softmax\left(\frac{QK^T}{\sqrt{d}}\right)V, \tag{20}$$

where x_u^s denotes the learner's cross-behavior fusion representation, and $softmax(\cdot)$ is the activation function. Then, we treat cognitive state feature x_u as a teacher signal to align the learner representation $\widetilde{x}_u$. This process is formalized by minimizing the following cross-task cognitive distillation loss function L_{CCD}:

$$L_{CCD}\left(x_u, \widetilde{x}_u\right) = \frac{1}{2}\left\|x_u - \widetilde{x}_u\right\|_2. \tag{21}$$

Next, to address the gap between x_u and $\widetilde{x}_u$ from two module, we introduce an adaptive gate mechanism [49] for dynamic feature fusion, generating a unified learner representation $\widetilde{x}_u$ that integrates cognitive states. The formulation is as follows:

$$x'_u = g \odot x_u + (1-g)\,\widetilde{x}_u, \tag{22}$$

where the gating unit g is obtained via a fully connected layer applied to h_s and $\widetilde{x}_u$:

$$g = sigmoid(W_6\,(h_s||\widetilde{x}_u) + b_6), \tag{23}$$

where W_6 and b_6 are trainable parameters. Finally, to construct a comprehensive learner representation reflecting learning demands, we employ a residual connection mechanism to linearly aggregate the learner's cognitive state feature x_u^s and cross-behavior fusion representation x'_u:

$$\check{x}_u = x'_u + x_u^s, \tag{24}$$

where $\check{x}_u$ denotes the learner feature fused with cognitive states and course preferences.

4.3 Performance Prediction

For final course recommendations, we compute recommendation scores through an inner product decoder [31]:

$$y_{u,c} = \check{x}_u \cdot x'_c. \tag{25}$$

To optimize the recommendation task, we adopt the Bayesian Personalized Ranking (BPR) loss [32]:

$$L_{BPR} = \textstyle\sum_{(u,c,c')\in D} -ln\ \sigma\left(y_{u,c} - y_{u,c'}\right), \tag{26}$$

where $D = (u, c, c')|r_{uc} = 1, r_{uc'} = 0$ represents the training data, $\sigma(\cdot)$ is the sigmoid function,$r_{uc} = 1$ indicates an observed interaction between learner and course, and $r_{uc'} = 0$ denotes a negative sample. The final loss for the recommendation task is defined as:

$$L = L_{BPR} + \lambda L_{CCD}, \tag{27}$$

where λ is a hyperparameter balancing the two loss functions.

5 Experiments

In this section, we conduct extensive experiments on real-world datasets to address the following research questions:

- **RQ1**: Does C^{3}Rec outperform general recommendation methods and existing course-specific approaches?
- **RQ2**: How do different components of the model affect the performance of the course recommendation task?
- **RQ3**: How do key hyperparameters influence the overall performance?
- **RQ4**: Is C^{3}Rec efficient for practical use?

5.1 Experimental settings

5.1.1 Datasets. To address our research questions, we evaluate the model on two public educational datasets: **MOOCCubeX** [44] and **EdNet** [6]. MOOCCubeX is sourced from XuetangX, a leading Chinese MOOC platform, offering over 3,000 high-quality courses from top institutions like Tsinghua University and Stanford University. Extract interaction logs from the entire year of 2020, excluding learners with insufficient historical interactions or abnormal behaviors. Due to the overly fragmented distribution of knowledge concepts in MOOCCubeX, we aggregated knowledge concepts with similar semantic representations such as "C Language" and "C Programming". EdNet is collected from Santa, an intelligent education platform focusing on TOEIC test preparation. We extract interaction logs from the entire year of 2019, excluding learners with fewer than 5 interactions or abnormal behaviors, forming a benchmark set of 1,580 learners, 899 courses, 11,540 exercises, and 300 knowledge concepts. Both datasets include learner-course and learner-exercise interactions, satisfying cross-task experimental requirements. Detailed dataset statistics are shown in Table 1.

For data splitting, following the **leave-one-out strategy** [22, 23], we chronologically order the interactions by enrollment date and divide the sorted dataset into training, validation, and test sets in the ratio of 8:1:1.

Table 1: Dataset Statistics

Dataset	MOOCCubeX	Ednet
Learners	10,930	1,580
Courses	289	899
Exercises	38,414	11,540
Knowledge Concepts	66	300
Learner-Course Interactions	25,189	16,051
Learner-Exercise Interactions	714,993	656,398

5.1.2 Evaluation Metrics. For each learner, each method generates a top-N course recommendation list for performance evaluation. To ensure evaluation reliability, we adopt two widely-used metrics: Recall@N [11] and [19], with N set to 10 and 20. Recall@N measures the proportion of ground-truth interacted courses covered in the recommendation list, while NDCG@N evaluates the ranking quality of recommendations. Higher values for both metrics indicate superior performance in recall rate and ranking rationality.

5.1.3 Baselines. To evaluate the effectiveness and superiority of C^{3}Rec, we compare our model with two categories of recommendation approaches: (1) **General recommendation methods:** MF, ENMF, LightGCN, SRGNN, GCSAN, SGL (2) **Course-specific recommendation methods:** HRL, KMCR, PCGNN.

- **MF** [21] represents recommendation scores as the product of user and item matrices to predict ratings for unobserved user-item interactions.
- **ENMF** [5] optimizes each element of user and item latent vectors to model implicit data, accelerating training efficiency.

- **LightGCN** [16] simplifies traditional GNNs by removing nonlinear transformations during message propagation to enhance recommendation efficiency.
- **SRGNN** [40] models user interaction sequences as graph-structured data, capturing dynamic item transitions via GNNs to profile user preferences.
- **GCSAN** [42] leverages global context repositories from interaction sequences and graph embeddings to alleviate data sparsity.
- **SGL** [39] enhances recommendation accuracy via self-supervised graph learning through edge/node dropout and random walk-based graph augmentation.
- **HRL** [46] employs hierarchical reinforcement learning to model learner representations, capturing multi-course interests via attention mechanisms.
- **PCGNN** [34] constructs a course prerequisite graph reflecting pedagogical principles and designs a gated GNN for course-level interest encoding.
- **KMCR** [2] integrates knowledge tracing with course recommendation, extracting learner traits from knowledge states to improve recommendations.

5.1.4 Implement details. We evaluate C^{3}Rec using Python 3.11.9 and PyTorch 2.4.1 under non-distributed training. All experiments are conducted on a Windows machine equipped with two 3080 GPUs. For fair comparison, hyperparameters of all baselines are configured based on their original papers or optimized through empirical experiments. For our model, hyperparameters are selected via grid search: the embedding dimension is 128, the number of graph convolution layers is set to 3, and the knowledge distillation loss weight is $5.00e \times 10^{-5}$. Additionally, we set the batch size to 128.

5.2 Overall Performance comparison (RQ1)

To validate the superiority and effectiveness of C^{3}Rec, we compare its recommendation performance against all baseline methods by using Recall and NDCG metrics. Experimental results are shown in Table 2.

From the experimental results, we observe that C^{3}Rec achieves the best performance across all compared methods and demonstrates significant advantages over existing state-of-the-art baselines. The superior performance of C^{3}Rec can be attributed to two key design aspects: 1) C^{3}Rec jointly models cognitive states from both course and exercise perspectives, enabling fine-grained recommendation enhancement based on knowledge states; 2) C^{3}Rec effectively integrates cognitive states with learning preference features through cognitive representation transfer. Meanwhile, we note that GCN-based methods, such as LightGCN and SGL, demonstrate strong performance across metrics, validating the effectiveness of mining implicit learning needs from the course interaction graph. Additionally, course-specific recommendation methods, such as KMCR and PCGNN, generally outperform general recommendation methods, highlighting the importance of domain-specific knowledge modeling. Notably, KMCR, which incorporates both knowledge states and personality traits, surpasses single-granularity modeling approaches in most metrics, further confirming the benefits of multi-dimensional learner profiling for recommendation performance.

5.3 Ablation study (RQ2)

In this subsection, we conduct ablation studies to validate the importance of each component in C^{3}Rec. We systematically eliminate or replace each component while keeping the remaining components unchanged, thereby yielding three variants:

- **C^{3}Rec w/o. GL:** Replaces the dual-granularity graph and cross-learning behavior graph with a single course interaction graph..
- **C^{3}Rec w/o. KD:** Removes the knowledge distillation module, directly concatenating cognitive proficiency representations from the diagnosis task with recommendation task embeddings.
- **C^{3}Rec w/o. CD:** Removes the cognitive diagnosis module, retaining only the course recommendation components.

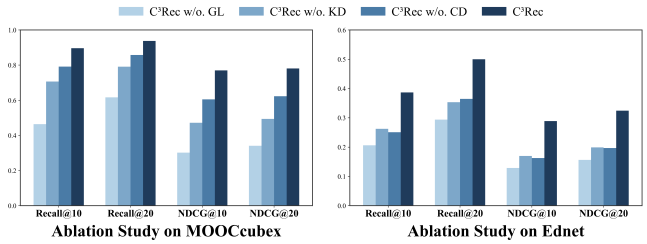


Figure 3: The result of ablation study.

Experimental results for these variants are shown in Figure 3. Overall, removing any component degrades recommendation performance, demonstrating their critical roles in modeling course preferences. Notably, C^{3}Rec w/o. GL exhibits significant performance drops, highlighting the effectiveness of our dual-granularity cross-behavior modeling in enhancing recommendations. Interestingly, we observe that the performance degradation rates of C^{3}Rec w/o. KD and C^{3}Rec w/o. CD differ across datasets. C^{3}Rec w/o. CD achieves better performance than C^{3}Rec w/o. KD on MOOCCubeX. This may occur because the average number of course interactions per learner on MOOCCubeX is only 2.305, indicating extreme sparsity. Without the cognitive diagnosis module, the sparse interaction data fails to support reliable preference modeling. Meanwhile, the average number of exercise interactions per learner on MOOCCubeX is 65.416. Cognitive states information mined from learners' exercise interactions via the cognitive diagnosis module enables cold-start course recommendations, effectively alleviating data sparsity issues.

5.4 Hyperparameter sensitivity (RQ3)

We further investigate the model's performance under different hyperparameters, including the embedding size d, the number of graph convolution layers l, and the distillation loss weight λ. Due to space limitations, we omit other hyperparameters with minor impacts on performance and primarily analyze these parameters on the MOOCCubeX dataset. The results are illustrated in Figure 4.

Table 2: Performance comparison of all methods on course recommendation task. The best results are highlighted in bold; the best baseline results are underlined.

Methods	MOOCCubeX				Ednet			
	Recall@10	Recall@20	NDCG@10	NDCG@20	Recall@10	Recall@20	NDCG@10	NDCG@20
MF	0.1819	0.2315	0.1333	0.1488	0.0797	0.1289	0.0587	0.0638
ENMF	0.3746	0.5080	0.3746	0.5080	0.0666	0.106	0.0449	0.0559
LightGCN	0.4932	0.6068	0.3402	0.3688	0.2994	0.3926	0.2012	0.2274
SRGNN	0.4871	0.6086	0.3062	0.3369	0.214	0.3092	0.1267	0.1497
GCSAN	0.4970	0.6272	0.2868	0.3196	0.2276	0.3443	0.1094	0.1383
SGL	0.6536	0.7483	0.449	0.4732	0.2901	0.3857	0.2009	0.2276
HRL	0.2088	0.3686	0.0936	0.1337	0.0678	0.1195	0.03283	0.0458
PCGNN	0.5769	0.6917	0.3965	0.4254	0.1864	0.2927	0.1065	0.1332
KMCR	0.5936	0.7714	0.4708	0.5153	0.3016	0.4301	0.1547	0.2095
C^{3}Rec	**0.8963**	**0.9374**	**0.7698**	**0.7809**	**0.3868**	**0.4999**	**0.2888**	**0.3243**

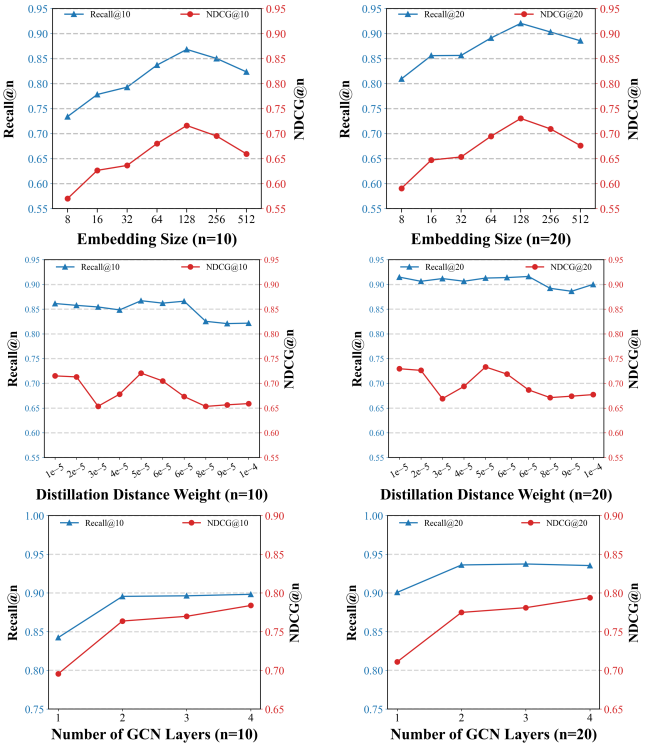


Figure 4: The result of hyperparameter sensitivity.

Embedding size: The embedding dimension significantly impacts model performance. We empirically set the dimension to [8, 16, 32, 64, 128, 256] and illustrate performance trends in Figure 4. As shown, C^{3}Rec's performance improves with increasing dimensions, peaking at d=128. Further increases degrade performance, likely due to overly sparse representations caused by high dimensionality.

Number of graph convolution layers: The number of graph convolution layers determines the model's ability to capture information from the graph structure. We vary the number of layers from 1 to 4, and as shown in Figure 4, the model achieves optimal performance when the number of layers is 3. This observation indicates that moderately increasing the number of layers helps capture collaborative filtering signals in the cross-behavior graph, but excessively high numbers may introduce additional noise.

Distillation loss weight: Distillation loss weight λ controls the influence of cognitive proficiency representations on the recommendation task. As shown in Figure 4, overly small λ hinders knowledge transfer from the diagnosis module, while excessively large λ overshadows cross-behavior learning and distracts the model from minimizing recommendation loss. Thus, $\lambda = \mathbf{5.00e \times 10^{-5}}$ achieves the best balance.

5.5 Case Study (RQ4)

In this part, we conduct one case to demonstrate the effectiveness of our method C^{3}Rec. We randomly sample a learner in the MOOCCubeX dataset, and showcase his/her course interactions, exercise interactions, and two top-5 recommendation lists from C^{3}Rec and other CR method. As shown in Figure 5, C^{3}Rec can provide better recommendation results that meet the learner's learning needs by considering his/her performance in exercises to capture cognitive states. From this learner's course interaction logs, it is evident that he/she performs well in Mechanics and Dynamics but struggles in more advanced Theoretical Mechanics. Therefore, our model recommends advanced university physics courses that also focus on fundamental content, while the other method's recommendation results have a weak correlation with learners' historical learning performance.

6 Conclusion

In this paper, we design a Cognitive Distillation-enhanced Cross-Behavior Course Recommendation model (C^{3}Rec) aimed to enhance online course recommendations by integrating cognitive diagnosis and knowledge distillation. Specifically, we introduce a dual-granularity cognitive diagnosis module to capture learners' cognitive states from exercise performance, and a cross-behavior course recommendation module to jointly profile learning preferences from two temporal interleaved learning behaviors. To bridge the gap between two tasks, we introduce a triple-stage joint learning

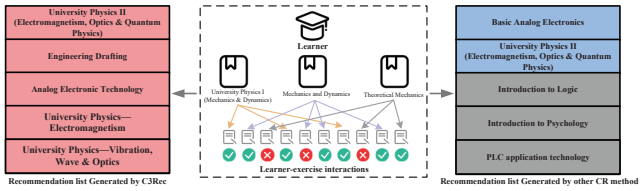


Figure 5: The case study of C^{3}Rec. Our method recommends courses that meet the learning needs of the learner based on their historical course interactions and exercise responses, while the recommendation results of the other method do not align well with the learner's historical learning record.

cognitive distillation to enhance course recommendation through the representation of cognitive states. Experiments on two real-world educational datasets demonstrate that our method outperforms existing approaches in Recall and NDCG metrics.

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